

Sahar Iqbal Malik

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PROFESSIONAL SUMMARY

Final-year Software Engineering student at FAST-NUCES (graduating 2026) with proven experience in AI/ML, full-stack development, and product analytics. Built Aegis – an enterprise security platform featuring a RAG chatbot for threat intelligence, plus ML-based intrusion detection. Product Analyst background driving 15% retention uplift. Strong systems thinker who turns complex workflows into clean, automated solutions. Seeking AI Engineering, Full-Stack Development, or Software Engineering roles.

EDUCATION

FAST-NUCES, Islamabad, Pakistan

June, 2026

BSC in Software Engineering

Relevant Coursework: (optional)

Westminster Academy, Islamabad, Pakistan

2022

SAT Score: 1360

WORK EXPERIENCE

Nayatel, *Product Management Intern*, Islamabad, Pakistan

June 2025 – July 2025

- Built full-stack D2C MVP in under 7 days using React + Firebase
- Proposed 3 AI-powered features to improve product adoption
- Researched 10+ AI platforms to guide product roadmap
- Conducted UX audit and delivered GTM strategy

CodingKey, *Product Analyst*, Islamabad, Pakistan

Jan 2024 – Aug 2024

- Improved product feature performance, contributing to ~15% retention uplift
- Performed data analysis using SQL and Python to guide product decisions
- Translated business needs into technical specifications
- Coordinated between engineering and design teams to deliver on deadlines
- Attended and conducted meetings with foreign clients

Hooked, *Founder*, Remote

June 2023 – Present

- Built and scaled business to 30-50 monthly orders
- Applied systems thinking to optimize inventory, customer support, and order tracking
- Managed end-to-end product lifecycle from design to customer retention

PROJECTS

Aegis IDS – ML-Powered Intrusion Detection System

React, FastAPI, XGBoost, CNN-LSTM, OpenAI API, LangChain, Vector DB, WebSockets

- Built enterprise-grade security platform combining real-time threat detection + AI-powered chatbot
- RAG Chatbot Feature: Security analysts can ask natural language questions (e.g., "What are the latest DDoS attack patterns?") and get instant answers from ingested security documents
- Multi-step RAG workflow: document ingestion → chunking → embedding → vector retrieval → LLM generation
- Integrated XGBoost and CNN-LSTM models to detect DDoS, port scans, and brute-force attacks in real-time
- Built React dashboard with live alerts, analytics, and SHAP-based explainable AI
- WebSocket streaming for real-time threat updates

Secure E2EE Messaging & File-Sharing Platform

React, Vite, Web Crypto API, IndexedDB, [Node.js](#), Express, Mongoose, MongoDB, JWT

- Built a full-stack end-to-end encrypted messaging and file-sharing system with a server-side blind relay model.
- Implemented AES-256-GCM encryption with authenticated ECDH (P-256) key exchange for secure session establishment.
- Added ECDSA signature verification to protect against tampering and man-in-the-middle attacks.
- Enforced replay protection using nonce validation, timestamp checks, and sequence-number tracking.
- Integrated JWT-based authentication, rate limiting, and audit logging for secure access and traceability.

Custom CNN – Pneumonia Detection

TensorFlow, Keras, Python, scikit-learn

- Developed chest X-ray classification CNN achieving 85.4% accuracy, 0.916 AUC
- Conducted ablation experiments with augmentation, dropout, and L2 regularization

AI Disaster Volunteer Allocation

Python, Streamlit, PuLP (Optimization), FastAPI

- Built optimization engine that allocates volunteers across zones based on severity and capacity
- Exposed API endpoints for structured task input and natural language queries
- Created interactive Streamlit dashboard with map visualization

CineMatch – AI Recommendation Engine

Python, NLP, scikit-learn, Genetic Algorithms

- Built recommendation system using sentiment analysis + evolutionary optimization
- Processed unstructured text data for personalized results

Iqbal Cardiocare – Clinic Management Platform

React, Node.js, MongoDB, Tailwind CSS, JWT

- Built a full-stack healthcare management platform for a cardiology clinic to streamline patient appointment
- Integrated email notifications for appointment confirmations and reminders
- Impact: Reduced manual appointment conflicts and improved patient intake efficiency

SKILLS AND INTERESTS

- *AI Agents & LLMs:* RAG Agents, LangChain, Vector Databases, Prompt Engineering, OpenAI API, Chatbot Development, XGBoost, CNN, TensorFlow, scikit-learn, NLP
- *Frontend:* React.js, Next.js, TypeScript, Tailwind CSS, shadcn/ui, Framer Motion, Zustand
- *Backend:* FastAPI, Node.js, Express.js, Python, WebSockets, REST APIs, JWT Authentication
- *Databases:* MongoDB, PostgreSQL, MySQL, Firebase
- *DevOps & Tools:* Docker, Vercel, Netlify, Git, GitHub, Postman
- *Core Concepts:* Systems Thinking, Agile/Scrum, Data Analysis (SQL/Python), Documentation, Clean Code
- *Soft Skills:* Systems Thinking, Documentation, Cross-Functional Communication, Problem-Solving

LEADERSHIP & EXTRACURRICULAR EXPERIENCE

FAST NUCES, Vice Head Volleyball, Islamabad, Pakistan

Aug 2025 - Dec 2025

- Led a 10+ member team, fostering team coordination, discipline, and a culture of mutual respect
- Organized practice schedules, team tryouts, and inter-departmental matches
- Acted as primary point of contact between team and university sports department
- Motivated team members through regular feedback sessions, improving overall team morale and performance

FAST Sustainable Development Society, Vice President Management, Islamabad, Pakistan

Aug 2024 – Dec 2024

- Led a team of 20+ members, delegating tasks and ensuring smooth execution of society events
- Organized BingX event at FAST-NUCES, managing logistics, budgets, and cross-team coordination
- Represented the society to external partners, sponsors, and university administration
- Developed operational SOPs for event management, improving efficiency for future teams

NASCON, *Marketing Team*, Islamabad, Pakistan

Dates

- Contributed to organizing one of Pakistan's largest student-led tech and sports fests
- Collaborated with a team of 20+ members to ensure seamless event execution
- Created promotional content and engaged with attendees from across Pakistan
- Enhanced communication and interpersonal skills through direct participant interaction